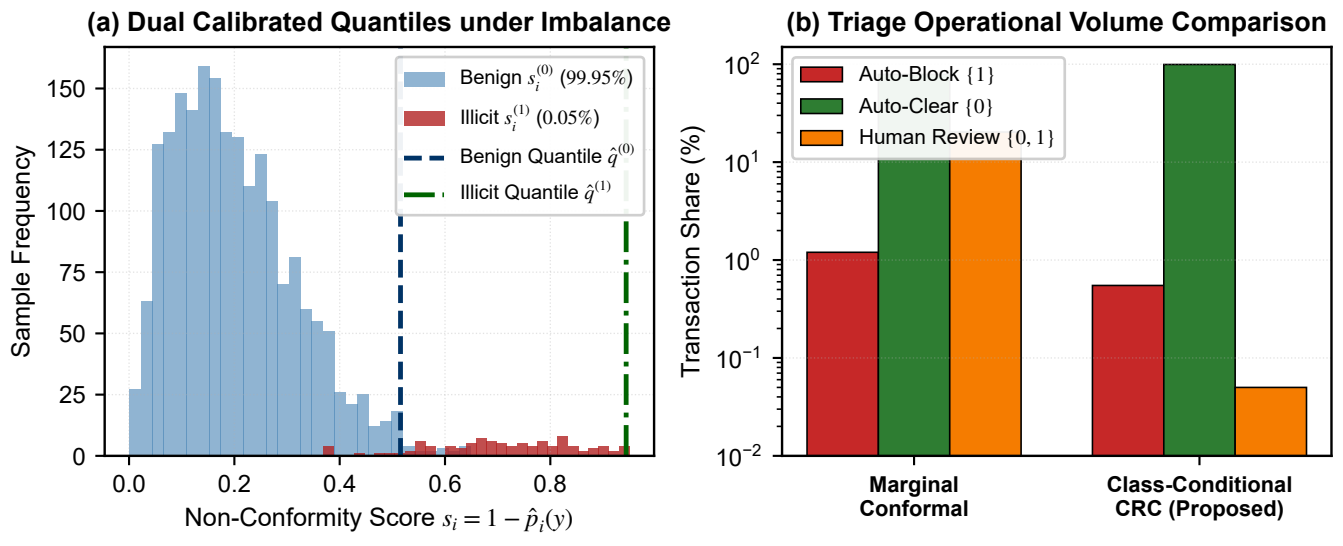

(a) Dual Calibrated Quantiles under Imbalance
Benign $s_i^{(0)}$ (99.95%)
Illicit $s_i^{(1)}$ (0.05%)
Benign Quantile $\hat{q}^{(0)}$
Illicit Quantile $\hat{q}^{(1)}$
Sample Frequency
Non-Conformity Score $s_i = 1 - \hat{p}_i(y)$
0
25
50
75
100
125
150
0.0
0.2
0.4
0.6
0.8
(b) Triage Operational Volume Comparison
Auto-Block {1}
Auto-Clear {0}
Human Review {0, 1}
Transaction Share (%)
$10^{2}$
$10^{1}$
$10^{0}$
$10^{-1}$
$10^{-2}$
Marginal Conformal
Class-Conditional CRC (Proposed)